Federated Learning on PathMNIST

A controlled comparison of FedAvg and FedProx under non-IID data and client label noise

Objective and setup

Five clients are formed from the official PathMNIST training split using a seeded Dirichlet partition (α=0.3). Client 0 receives 40% random label corruption, with every replacement label forced to differ from its original. Official validation data are partitioned into clean, client-specific held-out sets; the official test split is reserved for global evaluation.

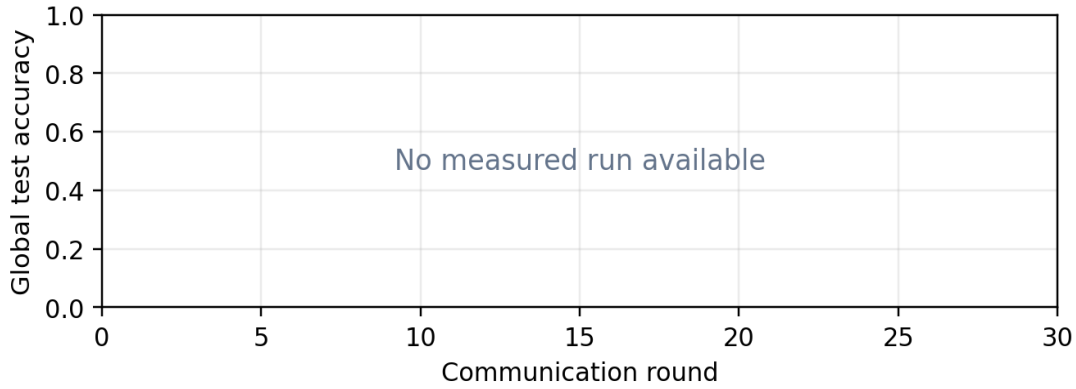
FedAvg averages client model parameters weighted by local sample count. FedProx uses the same local training procedure and adds μ/2■w–w_global■² with μ=0.01. Both start from identical weights and share the configured optimizer, rounds, local epochs and batch size.

Results pending

Training has not been run in this environment. Values below are intentionally pending; the configured command will replace them with measured results.

Measure	FedAvg	FedProx
Final global test accuracy	Pending	Pending
Final global test loss	Pending	Pending
Client 0 final validation accuracy	Pending	Pending
Client 1 final validation accuracy	Pending	Pending
Client 2 final validation accuracy	Pending	Pending
Client 3 final validation accuracy	Pending	Pending
Client 4 final validation accuracy	Pending	Pending

Global test accuracy by communication round



The axes are provided for context only; no accuracy observations are plotted.

Conclusion

This controlled setup is ready for execution, but the FedAvg/FedProx comparison and the effect of Client 0 label noise remain undetermined until measured runs are available.

Key settings

30 rounds · 2 local epochs · batch size 128 · SGD, lr 0.01, momentum 0.9 · FedProx μ=0.01 · seed 0 · 5 clients · Dirichlet α=0.3 · 40% noise on Client 0

Repository: not yet published. W&B; project: pathmnist-fedavg-fedprox (public link becomes available after authenticated upload and visibility change). See README for exact publishing steps.